

# Introduction. Number systems.

## Codes. Errors

### 1. Number systems:

- positional: each digit has an associated weight (binary, decimal, hexadecimal)
- non-positional: no weight associated (Roman numerals).

$$N = c_{m-1} \cdot b^{m-1} + c_{m-2} \cdot b^{m-2} + \dots + c_0 \cdot b^0 + c_{-1} \cdot b^{-1} + \dots$$

• The complement of a digit "c" is denoted  $\bar{c}$  in a given base b is:

$$\bar{c} = (b-1) - c.$$

ex.  $b_1 = 10$      $N = 365.6875$   
 $b_2 = 2$

$$\begin{array}{r|l} 365 & 2 \\ \hline \dots & \dots \\ \dots & \dots \\ \dots & \dots \end{array}$$

$$\begin{array}{r|l} 0 & 6875 \\ \hline 1 & 3750 \\ \text{①} & 7500 \\ 1 & 5000 \\ 1 & 0000 \\ \hline \end{array}$$

=> fractional (N)  
= 1011

